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DEALER BEST PRACTICE SERIES

Proper Lighting Provides Effective Working Environment

Application | Maintenance | Site Management | **Component Rebuild** | Component Life Management | Safety

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1.0 Introduction

Component Rebuild Center (CRC) work bay lighting is critical to several facets of CRC operations. Proper lighting provides an effective working environment for rebuild technicians by allowing them to see the smallest details, improving parts reuse decisions and workmanship in general. Good lighting also reduces eyestrain and worker fatigue, while promoting high professionalism to customers and employees. A clean, well-lit environment promotes quality and leads the effort in setting an internal image of excellence for all the employees to follow in their daily tasks.

Part reuse decisions are constantly being performed in a CRC. Parts are being washed or otherwise salvaged. Work is being performed above, below, and often inside small and large parts/components. Improper lighting in any of these scenarios can contribute to technician errors in judgment or misread scales. Installing unreusable parts, misadjusting systems, or missing process details may result in early component failure, short component service life, or possibly catastrophic failure. Substandard lighting limits technician capability and physically promotes a substandard capability to the customers and employees. Good shop lighting shows that the dealer is serious about promoting the quality advertised and sell.

2.0 Best Practice Description

Traditional shop lighting efforts are generally based on ceiling lights. A new shop will have more shadows and a generally darker environment with only ceiling lights. Lighting deficiencies often go unnoticed given the many adjustments needed to assure work gets performed effectively. As the shop's walls, ceilings, floors, equipment, and light fixtures get dirtier (especially with unprotected concrete), they reflect less light, creating a darker shop. This becomes more evident during the winter months with shorter days. Shops with normally open doors (not recommended) will also be darker during colder months when the doors remain closed during the day. Lighting is often improved by increasing the number of ceiling lights or random light fixtures in the shop. These additions help but are generally ineffective given their height and/or location. It's difficult to get enough ceiling lights to provide enough light with gray containers, equipment, and product, as well as dirty floors/walls absorbing much of it. Roof skylights help, but only during the first shift on sunny days.

Enhanced CRC lighting provides a well-lit environment without excessive lighting costs or maintenance. Effective lighting is provided through a comprehensive effort of coverage: ignoring one area will reduce lighting efficiency overall and provide inconsistent results. Increasing efforts in other area(s) may not provide enough to compensate for the loss.

- Ceiling lights are needed to provide enough <u>general</u> lighting for well-light aisles and bay work areas. Most dealers require lighting engineers to assure enough lighting given a ceiling height and shop colors. Roof-mounted skylights are highly recommended to provide natural lighting and reduce daytime lighting costs. Skylights must not replace ceiling fixtures due to lighting needed during 2nd and 3rd shifts and other times with marginal outside lighting.
- Sidelights are needed on walls and columns as needed to reduce shadows. Locate lights 10-12 ' above shop floor. These fixtures provide lighting for the work bay area, on the side and below major components, and workbench tops.

- Task/local lighting should be located above workbenches. These lights are typically 4' fluorescent lights suspended at the center of the workbench approximately 7' above the floor. These fixtures are primarily for workbench tasks and parts inspections. They also greatly supplement the sidelights since the workbenches are on the work bay perimeter and close to the major component.
- Droplights are needed on every work bay perimeter, on rewinding reels. Stationary lighting fixtures are not capable of directing enough light to tasks, such as work on the inside of part cavities or parts in the middle of the work bay. Droplights are best stored by suspending them from reels to assure:
    - The lights are easily retrieved for use.
    - Easily stored for cleanup/protection.
    - The technician does not substitute a lower wattage flashlight.
- Large bridge crane beams should have light fixtures to compensate as they often block ceiling lights when moved to a working position. This prevents reducing the light under the hoist being used.
- Use reflective colors in the shop to conserve available light and create a professional work environment. Smooth painted surfaces help prevent dirt/dust from sticking and are easier to clean.
    - Ceiling should have shiny or light painted surfaces – including structural beams/ trusses. Avoid exposed spray-on or paper-faced insulation as these surfaces absorb dirt and are difficult to clean without damage.
    - Walls are often painted tan or off-white up to 7'-8' high to reduce noticeable staining. The upper area of the wall should be painted white to reflect as much light as possible since it is much less vulnerable to staining.
    - Floors must be sealed (reapplied regularly) so oil/grease are not absorbed, dark staining is prevented, and cleaning efforts are reduced. Floors may be painted lighter than concrete (if impact damage is controlled) for maximum light reflection and cleaning ease.

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  - Paint equipment, especially cabinets, cranes, and workbenches, with bright yellow or another reflective color. Use high-gloss paint for maximum durability and cleaning ease.
- Keep the CRC clean to prolong its lighting effectiveness. Attractive paint and effective lighting deteriorate quickly (along with employee pride) if not kept clean. Focus on areas originating dirt, grease and airborne debris to control spreading contaminates. Maintain areas daily, especially disassembly/cleaning areas with:
  - Downdraft tables.
  - Contained blast cleaning.
  - Smooth painted walls.
  - Large oil-trapping grids in the floor.
  - Effective cabinet washer evacuation.
  - Utilities, such as air lines, on retracting reels.
  - Equipment/tool racks/cabinets to maintain organization.

3.0 Implementation Steps

- Assess the CRC's status and document conclusions in text and pictures. Focus on:
  - Lighting capability in every aisle, work bay, and corner. Observe shadows and transient conditions with crane usage.
  - Special lighting needs, such as for detailed inspections.
  - Lighting during all working shifts and with all doors closed.
  - Typical shop cleanliness conditions and cleaning habits.
  - Building and equipment colors. Note reflectivity, condition, and cleanliness.
- Establish lighting needs, placement, and painting requirements.
  - Research/establish lighting specialist as a consulting SME.
  - Establish lighting equipment suppliers and support as needed.
  - Research Caterpillar equipment recommendations for fixtures, arrangement, etc.
  - Determine painting resources and support services.
- Develop and design CRC color scheme surface requirements and paint types.
- Adjust and document housekeeping requirements, procedures, roles and responsibilities (with quality checklists as needed).
- Establish new repair/rebuild time requirement targets as needed.
- Present strategy to shop employees.
- Install the equipment changes into the shop (This may occur in phases as needed.)
- Train employees to use the new procedures/equipment. Follow-up/enforce immediately.
- Review process and equipment performance once established.
- Establish and install any adjustments as needed.

4.0 Benefits

- Reduced non-chargeable (general cleaning) labor - work bay cleanliness is maintained during normal daily activity, not at the “end of period cleanup”. Lighting better shows what should be cleaned. Saved labor is applied to increased revenue, providing more profits.
- Increased quality – more accurate parts reuse inspections and more consistent cleaning standards. Technicians provide effective workmanship due to less (eye and general) fatigue.

- Improved technician safety – obstacles are easier to see and maintaining cleanliness reduces airborne debris.
- Improved CRC image – customers see a more professional rebuild environment, helping to increase price realization. Employees experience a quality/professional environment, encouraging them to provide the same level of contribution in their efforts.

5.0 Resources Required

- Investment costs vary, depending on lighting-related gaps found and local labor/material costs:
  - Light fixture installation
  - New paint required on worn/abused equipment.
  - Preparation for building paint job(s)
  - Skylight installation.
- Support Equipment
  - Equipment organization aids (racks, cabinets).
  - Airborne debris containment (downdraft tables, evacuation systems, etc.)
  - Utility reels for effective cleaning efforts.
  - Used oil grids for oil containment.
- People
  - Establish current painting, and other environmental< concerns with team.
  - Establish process, facility, and equipment improvements through team.
- Training
  - Establish the improvement benefits with the production team.
  - Define, establish, and enforce the new process and duties with the production team.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

None.

8.0 Acknowledgements

This Best Practice was written by:

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